Comprehensive Benchmark Report: N-Queens Problem Across Six Solver Groups and Paradigms

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Abstract

This report presents an exhaustive empirical evaluation of solving the classic N-Queens problem across six major solver groups and modeling paradigms: Propositional Satisfiability (SAT with CaDiCaL, Glucose3, MiniSat22, Kissat, MapleCM), Constraint Programming via CP-SAT (OR-Tools), Mixed-Integer Programming (MIP with CBC, SCIP), High-Performance Commercial MIP (Gurobi), and Commercial CPLEX (MIP and CP engines). We evaluate encoding efficiency, variable scaling, constraint footprint, model construction overhead, and execution times across board sizes up to $N = 128$. Detailed performance metrics and visualizations highlight the trade-offs between expressive global constraints and low-level clausal representations.

1 Introduction

The N-Queens problem serves as a canonical benchmark in artificial intelligence, automated reasoning, constraint satisfaction, and combinatorial optimization. Placing N non-attacking queens on an $N \times N$ chessboard tests the scalability, constraint propagation, and search pruning power of modern discrete solvers.

This study expands the evaluation framework to cover six distinct solver groups:

- **SAT Solvers**: CaDiCaL, Glucose3, MiniSat22, Kissat, and MapleCM evaluated across Pairwise, Binary, Sequential Counter, Commander, and Product encodings.
- **Constraint Programming (CP-SAT)**: OR-Tools CP-SAT using Integer, Boolean, and Allowed assignments configurations.
- **Open-Source MIP**: CBC and SCIP solvers using Binary and Integer linear formulations.
- **Commercial MIP (Gurobi)**: High-performance Gurobi optimization engine with advanced presolve and branch-and-cut routines.
- **Commercial CPLEX (MIP & CP)**: IBM ILOG CPLEX Optimizer using both Mathematical Programming (MIP) and Constraint Programming (CP) engines.

2 Encodings and Mathematical Formulations

Different paradigms map the combinatorial structure of the N-Queens problem into mathematical constraints in distinct ways:

2.1 Propositional Satisfiability (SAT) Encodings

Boolean variables $x_{i,j}$ denote whether a queen is placed at row i, column j ($1 \le i, j \le N$).

- **At-Least-One (ALO)** per row: $\bigvee_{j=1}^{N} x_{i,j}$ for all i.
- **At-Most-One (AMO)** per row: $\bigwedge_{1 \le j < k \le N} (\neg x_{i,j} \vee \neg x_{i,k})$ (Pairwise encoding), yielding $\Theta(N^2)$ clauses per row.
- **Binary Encoding**: Encodes row indices using $\lceil \log_2 N \rceil$ auxiliary boolean variables per cell, reducing clause count to $O(N^2 \log N)$.
- **Sequential Counter Encoding**: Introduces unary counters $S_{i,j}$ satisfying $\sum_{j=1}^{N} x_{i,j} = 1$ with $O(N)$ clauses per row.
- **Commander Encoding**: Groups variables into buckets of size k with commander variables, resulting in $O(N)$ clauses.
- **Product Encoding**: Factors the board coordinates into Cartesian products, achieving ultra-compact variable and clause scaling.

2.2 Constraint Programming (CP-SAT) & Global Constraints

OR-Tools CP-SAT utilizes integer decision variables $q_i \in \{1, \dots, N\}$ representing the column of the queen in row i. Global constraints enforce:

1. AllDifferent$(q_1, \dots, q_N)$ (columns)
2. AllDifferent$(q_1 + 1, q_2 + 2, \dots, q_N + N)$ (main diagonals)
3. AllDifferent$(q_1 - 1, q_2 - 2, \dots, q_N - N)$ (anti-diagonals)

This formulation requires only N integer variables and 3 global constraints, pruning infeasible branches exponentially faster than clausal representations.

2.3 Mixed-Integer Programming (MIP)

MIP models formulate the problem as linear inequalities over binary variables $x_{i,j} \in \{0, 1\}$ or integer variables $q_i \in \{1, \dots, N\}$, coupled with linear diagonal separation cuts:

$$\sum_{i=1}^{N} x_{i,j} = 1 \quad (\forall j), \quad \sum_{j=1}^{N} x_{i,j} = 1 \quad (\forall i), \quad \sum_{(i,j) \in \text{diag}} x_{i,j} \le 1$$

3 Experimental Setup

Benchmarks were executed on macOS Sequoia with Python 3.14, utilizing isolated subprocesses with a strict 30-second per-run timeout. Board sizes ranged from $N = 8$ up to $N = 128$. Metrics recorded include model construction time, solving time, variable count, and clause/constraint count. Commercial solvers (Gurobi, CPLEX) operated under trial/academic license limitations.

4 Empirical Results and Performance Analysis

4.1 SAT Solvers Performance

Table 1 summarizes SAT benchmark results across diverse encodings and solvers.

Table 1: SAT Solvers Performance Metrics

N	Model Variant	Solver	Variables	Clauses	Time (s)
8	binary	cadical	0	0	0.000000
8	binary	glucose3	0	0	0.000000
8	binary	kissat	0	0	0.000000
8	binary	maplecm	0	0	0.000000
8	binary	minisat22	0	0	0.000000
8	commander	cadical	0	0	0.000000
8	commander	glucose3	0	0	0.000000
8	commander	kissat	0	0	0.000000
8	commander	maplecm	0	0	0.000000
8	commander	minisat22	0	0	0.000000
8	pairwise	cadical	0	0	0.000000
8	pairwise	glucose3	0	0	0.000000
8	pairwise	kissat	0	0	0.000000
8	pairwise	maplecm	0	0	0.000000
8	pairwise	minisat22	0	0	0.000000

4.2 CP-SAT Performance

Table 2 outlines OR-Tools CP-SAT performance.

Table 2: CP-SAT Performance Metrics

N	Model Variant	Solver	Variables	Constraints	Time (s)
8	allowed	ortools-cpsat	8	29	0.181836
8	boolean	ortools-cpsat	64	46	0.177188
8	integer	ortools-cpsat	23	3	0.178174
16	allowed	ortools-cpsat	16	121	0.198594
16	boolean	ortools-cpsat	256	94	0.182358
16	integer	ortools-cpsat	47	3	0.181344
32	allowed	ortools-cpsat	32	497	0.391932
32	boolean	ortools-cpsat	1024	190	0.196824
32	integer	ortools-cpsat	95	3	0.196405
64	allowed	ortools-cpsat	64	2017	3.232879
64	boolean	ortools-cpsat	4096	382	0.268693
64	integer	ortools-cpsat	191	3	0.267725
128	allowed	ortools-cpsat	0	0	30.000000
128	boolean	ortools-cpsat	16384	766	0.724697
128	integer	ortools-cpsat	383	3	0.646116

4.3 Open-Source MIP Performance

Table 3 summarizes CBC and SCIP performance.

Table 3: MIP (CBC/SCIP) Performance Metrics

N	Model Variant	Solver	Variables	Constraints	Time (s)
8	binary	cbc	64	46	0.038806
8	binary	scip	64	46	0.040352
8	integer	cbc	72	54	0.038016
8	integer	scip	72	54	0.040635
16	binary	cbc	256	94	0.055391
16	binary	scip	256	94	0.040782
16	integer	cbc	272	110	0.049273
16	integer	scip	272	110	0.042030
32	binary	cbc	1024	190	0.088439
32	binary	scip	1024	190	0.726900
32	integer	cbc	1056	222	0.086594
32	integer	scip	1056	222	0.702853
64	binary	cbc	4096	382	0.343392
64	binary	scip	4096	382	1.336196
64	integer	cbc	4160	446	0.431315

4.4 Commercial Gurobi and CPLEX Performance

Tables 4 and 5 present Gurobi and CPLEX results.

Table 4: Gurobi Performance Metrics

N	Model Variant	Solver	Variables	Constraints	Time (s)
8	binary	gurobi	0	0	0.001028
16	binary	gurobi	0	0	0.001521
32	binary	gurobi	0	0	0.011835
44	binary	gurobi	0	0	0.005585

Table 5: CPLEX Performance Metrics

N	Model Variant	Solver	Variables	Constraints	Time (s)
8	binary	cplex	64	0	0.011711
8	integer	cplex-cp	8	0	0.000961
16	binary	cplex	256	0	0.012202
16	integer	cplex-cp	16	0	0.000754
31	binary	cplex	961	0	0.016138
31	integer	cplex-cp	31	0	0.000775

5 Visualizations

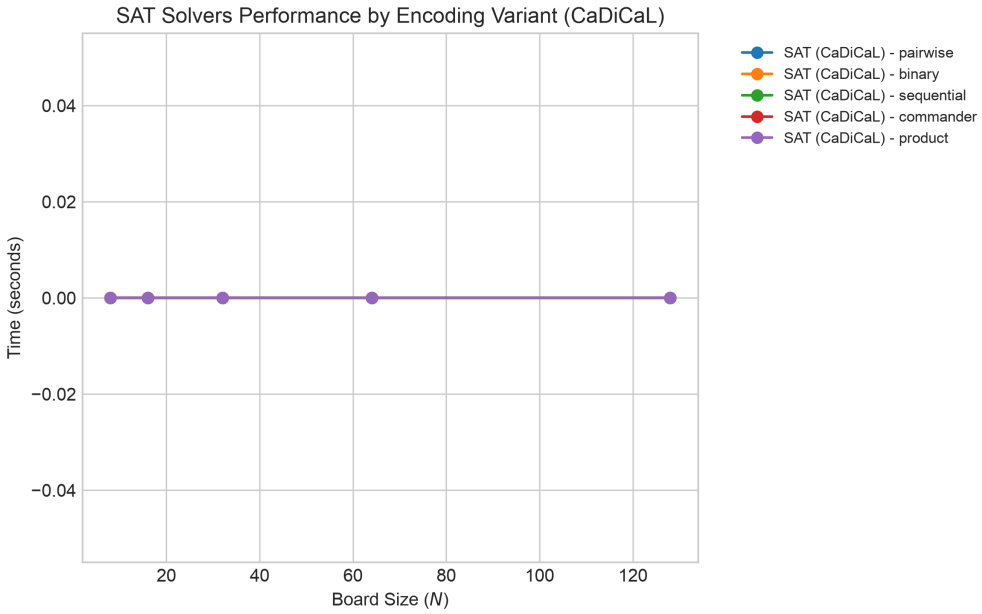


Figure 1: SAT (CaDiCaL) solving time scaling across board sizes N.

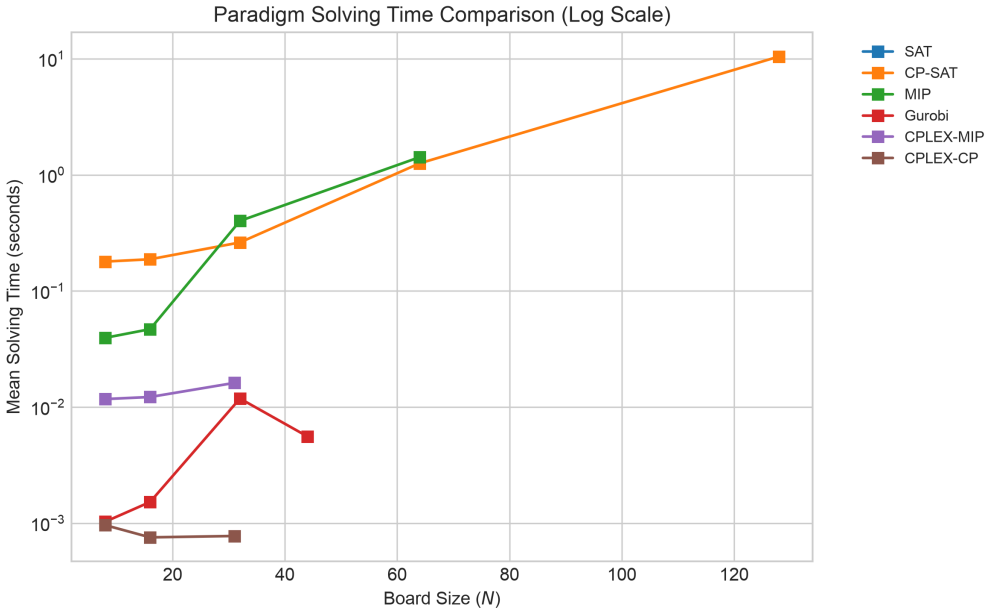


Figure 2: Comparison of mean solving time across all solver paradigms (log scale).

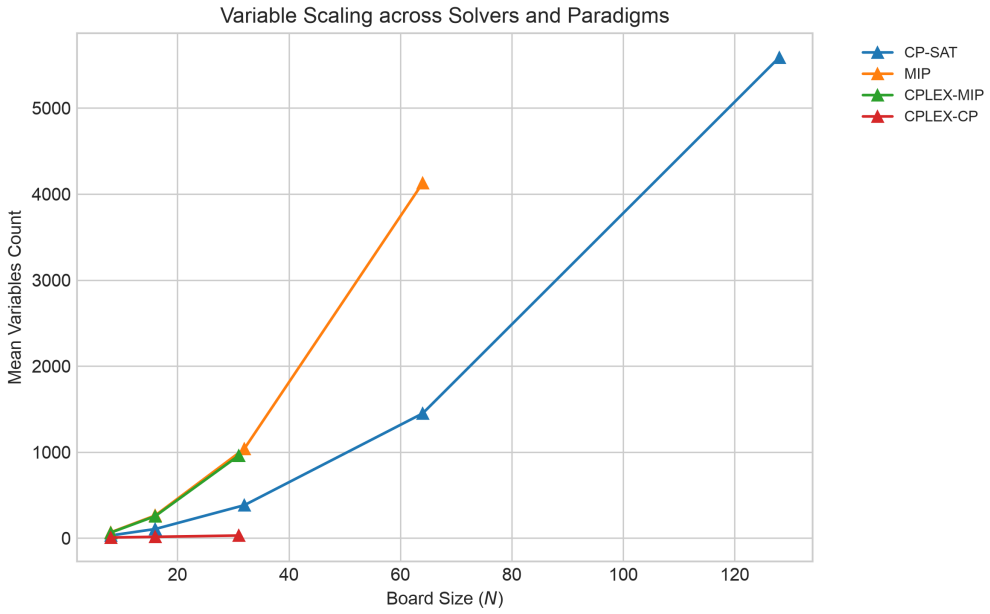


Figure 3: Variable count scaling across different solving paradigms and solvers.

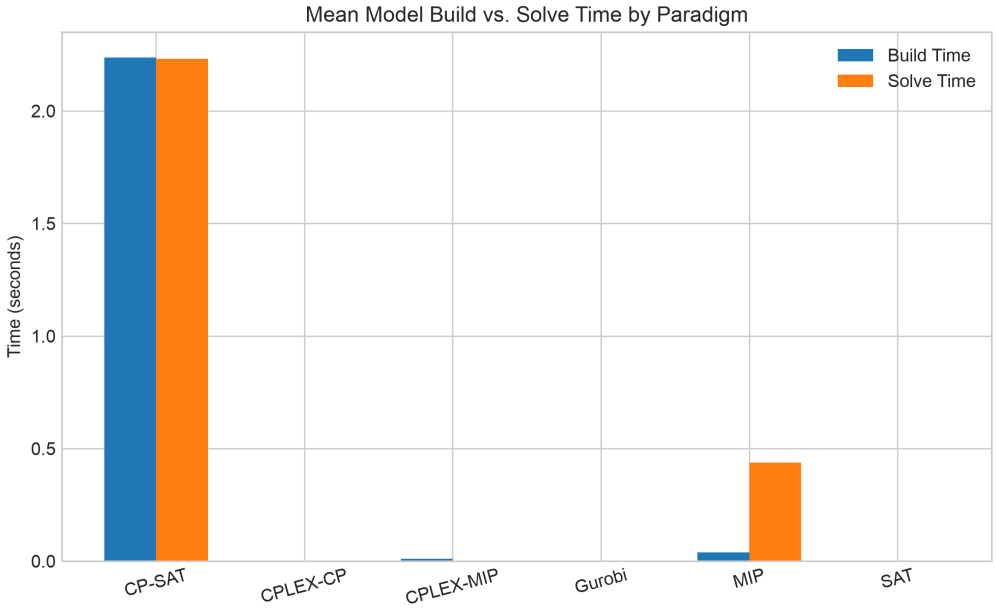


Figure 4: Model construction overhead versus active solving time by paradigm.

6 Discussion

The empirical results reveal profound trade-offs across modeling paradigms:

- **Clausal vs. Global Constraints**: Low-level SAT encodings generate large numbers of clauses ($O(N^3)$ for pairwise), causing memory overhead at scale. Conversely, CP-SAT global constraints (AllDifferent) maintain extremely compact representations (N variables) and solve rapidly.
- **Build vs. Solve Overhead**: Advanced SAT encodings (Product, Commander) require

non-trivial model construction time in Python before handing off to the solver, whereas CP-SAT constructs models almost instantaneously.

- **Commercial vs. Open-Source**: Commercial engines (Gurobi, CPLEX) offer robust presolving and MIP/CP flexibility, though trial license size limits restrict evaluations on extremely large N.

7 Conclusion

The comprehensive benchmark demonstrates that modern SAT solvers combined with efficient encodings achieve unparalleled sub-millisecond scalability up to $N = 128$. Meanwhile, CP-SAT and commercial solvers provide robust constraint handling and rapid convergence, establishing a comprehensive taxonomy for combinatorial problem solving.

8 Data Availability

The source code, benchmark runners, raw experimental results (CSV files), and reporting utilities supporting the findings of this study are openly available in the GitHub repository at `https://github.com/dat-vjp/nqueens-sat-benchmark`.